Diagnostic Scanner

Category: **Automotive** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Analyzer Machines; Electronic diagnostic devices

What it is used for

An automotive diagnostic scanner communicates with a vehicle's electronic control units to retrieve fault codes, display sensor data, and perform supported diagnostic or service functions. Correct vehicle identification, safe connection, accurate interpretation, and approved testing procedures help diagnose faults without damaging vehicle electronics.

Main parts

- Display screen presents diagnostic information.
- Control buttons or touchscreen navigate menus.
- Diagnostic connector cable links the scanner to the vehicle port.
- Vehicle communication interface exchanges data with electronic control units.
- Power supply or rechargeable battery powers the scanner.
- Software provides vehicle-specific diagnostic functions.
- USB or wireless interface supports updates or data transfer where fitted.
- Protective housing and cable connectors protect the device.

Before you start

1. Only trained personnel should perform vehicle diagnostics and service functions.
2. Confirm the vehicle make, model, year, powertrain type, and supported communication protocol.
3. Inspect the scanner, cable, connector pins, housing, and power supply for damage.
4. Read the scanner instructions and confirm that the software supports the target vehicle and intended function.
5. Park the vehicle safely, apply the parking brake, select the appropriate gear position, and follow vehicle-specific ignition instructions.
6. Check the vehicle battery condition; low voltage may interrupt communication or affect diagnostic results.
7. For electric and hybrid vehicles, follow high-voltage safety procedures and do not touch orange high-voltage cables or components unless qualified and authorised.

How to use

1. Switch on the scanner and select the correct vehicle identification details or approved automatic identification option.
2. Locate the vehicle's diagnostic port using the service information and inspect it for damage or contamination.
3. Connect the diagnostic cable securely to the vehicle port without forcing the connector.
4. Switch the vehicle ignition or diagnostic mode to the state required by the scanner instructions.

5. Select the required system, such as engine, transmission, ABS, airbag, or battery management, where supported.
6. Read and record stored, pending, and permanent diagnostic trouble codes as applicable.
7. Review freeze-frame data and live sensor values when available, comparing them with vehicle service specifications.
8. Perform supported active tests or service functions only when authorised and when their prerequisites are satisfied.
9. Do not clear fault codes until the original codes and relevant data have been recorded and the repair plan permits clearing.
10. After authorised repairs, clear codes if appropriate and repeat the scan or specified functional test.
11. Turn off the diagnostic session and disconnect the scanner carefully according to the instructions.
12. Save or document results, disconnect the device, and return the vehicle to a safe operating condition.

⚠ Safety

- Never perform coding, programming, actuator tests, or resets unless the function is supported and the procedure is understood.
- Do not disconnect the scanner or interrupt vehicle power during a programming operation.
- Use vehicle-specific procedures for airbag, ABS, immobiliser, and high-voltage system diagnostics.
- Keep cables clear of pedals, steering components, fans, belts, and hot engine parts.
- Do not assume a diagnostic trouble code identifies the failed component; verify the fault using service information and testing.
- Only qualified personnel should access high-voltage EV systems or perform safety-critical vehicle repairs.

Care & storage

- Keep scanner software and vehicle coverage updated using approved update procedures.
- Clean the housing and connectors with a manufacturer-approved method.
- Inspect cable insulation, connector pins, and battery condition regularly.
- Store the scanner in a dry, protected case away from excessive heat and moisture.
- Report recurring communication failures and arrange authorised repair if the device is damaged.

Common mistakes to avoid

- Selecting the wrong vehicle or control system.
- Replacing parts based only on a fault code without confirming the cause.
- Clearing codes before recording diagnostic information.
- Interrupting power or communication during programming or other critical service functions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.